# Shubhadeep Mandal

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## Education

**International Institute of Information Technology, Hyderabad** July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* *CGPA: 9.05*

## Research & Open-Source Experience

**Core Contributor, RookDB** Aug 2026 – Present
*Independent Study, DSAC Lab, IIIT Hyderabad | Rust, SQL, Databases* *Hyderabad, India*

- Co-designed the lab's open-source SQL database in Rust: storage engine, buffer pool, B+Tree indexes and query executor.
- Built the cost-based query planner and `EXPLAIN ANALYZE`, and used them to find and fix slow queries.
- Led performance work that made the benchmark suite **14x faster**; table scans run at **20M rows/s**.
- Main contributor to the engine, with **1,200+ tests** including differential testing against SQLite.

## Projects

**Peer-to-Peer Distributed File Sharing System** | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

**Negotiate, Then Solve: AI Scheduling Platform** | *Python, FastAPI, React, TypeScript, OR-Tools, LLMs*

- Built a full-stack scheduling assistant: an LLM turns requests from email, chat and the web into constraints for a solver.
- FastAPI backend with SQLite and a React frontend; it builds a **150-course** semester timetable with no rule violations.
- Fine-tuned a small 4B model with LoRA that outperformed the Gemini model it was trained to imitate.

**IndicLLM: Transformer Pretraining & Reasoning SFT** | *Python, PyTorch, Transformers*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Tested whether the models were reasoning or memorising answers before reporting results.

## Technical Skills

**Languages**: C/C++, Python, SQL, TypeScript, Rust, Bash.
**Backend & Web**: FastAPI, PostgreSQL, SQLite, React, Next.js.
**Machine Learning**: PyTorch, Transformers, LoRA, NumPy, Pandas.
**Systems & Tools**: Linux, Multithreading, Sockets, Git, Docker, GitHub Actions.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

## Interests

- Reading web novels and manga.
- Playing chess and badminton.